# Gyutae Oh

CURRICULUM VITAE

Media System Lab (MSL), Sungkyunkwan University
Electrical and Computer Engineering
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Portfolio
Google Scholar
Git
LinkedIn

AI graduate researcher focused on **efficient learning**, **continual learning**, **anomaly detection**, and **medical image analysis**. Currently pursuing a Ph.D. in Electrical and Computer Engineering at Sungkyunkwan University under the supervision of Prof. Jitae Shin.

## EDUCATION

**Sungkyunkwan University, Ph.D. in Electrical and Computer Engineering** — Feb 2026 – Present

**Sungkyunkwan University, M.S. in Electrical and Computer Engineering** — Feb 2024 – Feb 2026

GPA: 4.07/4.5

**Korea National University of Transportation, B.S. in Computer Engineering** — Mar 2019 – Feb 2024

GPA: 3.79/4.5

## RESEARCH EXPERIENCE

**Media System Lab (MSL), Sungkyunkwan University** — Jan 2024 – Present

Advisor: Jitae Shin

**Applied Machine Intelligence Lab (AMI-Lab), KNUT** — Mar 2022 – Jun 2023

Advisor: Jeonghwan Gwak

## MILITARY SERVICE

## RESEARCH INTERESTS

- Continual Learning
- Efficient Learning
- Anomaly Detection
- Medical Image Analysis
- PyTorch
- Evaluation

## RESEARCH OUTPUTS

**11 total papers**

4 preprints · 7 peer-reviewed publications

**1 patent**

7 scholarships · 4 awards · 1 certification · 1 service

## HONORS & AWARDS

**Hanium: Companion Dog Eye Disease Detection Assistant App — Selected Works Prize (2023.11)**

Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

### Republic of Korea Army, Sergeant, Honorably Discharged

2020 – 2022

TOD operator at Gachilbong GOP, 21st Infantry Division, under a direct unit of III Corps

## SELECTED PUBLICATIONS

† First author, * Corresponding author

**Routing Prompts Across Layers: A Global Memory Module for Prompt-Based Continual Learning**

**Gyutae Oh**, Jitae Shin

International Conference on Computer Research and Development (ICCRD), January 24, 2026

**Multimodal Fusion Framework Using Contrastive Learning for Exposure Keratopathy**

**G Oh**†, Y Won†, D Lim, J Shin*

International Workshop on Ophthalmic Medical Image Analysis (MICCAI 2025 Workshop), 2025

Paper Code

**Patch Core++: Augmenting Coreset for Superior Industrial Anomaly Detection**

**G Oh**, J Shin*

IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia), 2024

Paper

**Feature Ensemble-based Wolff Parkinson White Syndrome classification through ECG**

**G Oh**†, I Kim, B Kim, Y Jeon, J Gwak*

Proceedings of the Korean Society of Computer Information Conference, 2023

Paper

## UNDER REVIEW

### ECCV 2026

2 anonymous submissions under review

### MICCAI 2026

1 anonymous submission under review

### ICIP 2026

1 anonymous submission under review

### Companion Dog Eye Disease Detection Assistant App — Silver Prize & Graduation Project (2023.08)

Korea National University of Transportation

Video

### Hanium: Automated Serving Robot using Object Detection & SLAM — Bronze Prize (2020.08)

Hanium ICT Mentoring Program, hosted by the Ministry of Science and ICT (MSIT), Korea

Video

### Campus Contest: Automated Serving Robot using Object Detection & SLAM — Grand Prize (President's Award) (2020.09)

Korea National University of Transportation

Video

## SCHOLARSHIPS

### SKKU BK21 Fellowship (Specialized Scholarship) (2024)

Sungkyunkwan University

### University Innovation — Merit & Living Expense Scholarship (2024)

Sungkyunkwan University

### Academic Excellence Scholarship (awarded 5 times, 2019–2023)

## SELECTED PROJECTS

### Exposure Keratitis Diagnosis & Severity Prediction (Year 2)

**2025.10 – Present**

Sungkyunkwan University · Kangbuk Samsung Hospital collaboration

Extending multimodal medical AI research toward model compression and clinically practical deployment.

### AI CHAMPION

**2025.07 – 2025.09**

Fundus-image-based multi-disease screening project with cross-lab and hospital collaboration; selected as a Top 100 team out of approximately 500 teams.

### Exposure Keratitis Diagnosis & Severity Prediction (Year 1)

**2024.10 – 2025.10**

Developed multimodal AI models for exposure keratitis diagnosis and severity prediction from anterior segment photographs, leading to a workshop publication and a domestic patent application.

Korea National University of Transportation

## PATENT

### Device and Method for Diagnosing Exposure Keratopathy based on Multimodal Fusion Learning

Application No. 10-2025-0121342 · Filed Aug 28, 2025

Two-stage multimodal fusion learning framework for automatic grading of Exposure Keratopathy (EK). Trained with four ocular imaging modalities collected at a tertiary hospital (broadbeam, slit-beam, scattering, and blue-light): modality-wise features are aligned and fused during training, while inference enables modality-agnostic classification using only a single broadbeam image to implicitly leverage information from other modalities.

## CERTIFICATION

### Advanced Data Analytics Semi-Professional (ADsP)

Issued by Korea Data Agency (KDATA) · Sep 2025

**ID**: ADsP-046012145

## PROFESSIONAL SERVICE

### Reviewer (2026)

IEEE International Conference on Image Processing (ICIP)

Updated from portfolio content on March 19, 2026.